



NQRUST-ANALYTICS

Ask in Plain Language | Answer with Trustworthy Data

A self-hosted AI-powered Business Intelligence platform that answers business questions from two sources at once — your own data warehouse (via auto-generated SQL) and your uploaded knowledge documents (via document RAG). One chat, two sources, every answer cited.

NL→SQL

Text-to-SQL

RAG

Chat-to-Docs

11+

DB
Connectors



Bilingual
ID / EN

Version 2.0 - Brochure
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INTRODUCTION

What Is NQRust-Analytics?

A self-hosted AI-powered Business Intelligence platform that turns natural-language questions into SQL, charts, and document-grounded answers — fully on your infrastructure.

NQRust-Analytics answers business questions from **two data sources at once** — **your data warehouse** (via auto-generated SQL) and **your uploaded knowledge documents** (PDFs, Word docs, Markdown, slides). Built on a Rust query engine with MDL semantic layer and vector retrieval, the platform classifies each question, routes it to the right pipeline, and returns SQL results, charts, and document-cited narratives — all in one chat.

DATABASE + DOCUMENTS. ONE CHAT.

Two data sources, one conversation — the platform routes each question automatically.



Chat-to-Data

Type a business question in English or Indonesian. The system classifies intent, generates SQL against your schema, validates and self-corrects up to three times, then returns the answer with an auto-picked chart and a written narrative.



Knowledge Documents

Upload your documents — PDF, Markdown, DOCX, or PPTX — and ask questions about them. Answers come grounded in the document with inline page-level citations. Tesseract OCR for scanned PDFs in Indonesian and English.



Dashboards & Workspace

Pin chart results into shareable dashboards. Thread-based chat history captures every question, SQL, and follow-up. Knowledge management lets your team curate SQL pairs and instructions to steer the LLM.

SUPPORTED DATABASE CONNECTORS

Connect to your existing data warehouse — read-only, no migration required.

PostgreSQL

MySQL

BigQuery

Snowflake

Redshift

DuckDB

ClickHouse

Trino

Athena

Oracle

SQL Server

+ more

DEPLOYMENT



Docker Compose

Single-host stack orchestrated by Docker Compose. UI, AI service, query engine, vector store, and workspace database — all containers managed from one file. Production deployment in minutes.



Choose Your LLM Provider

The platform comes pre-integrated with multiple **external LLM services**. Pick which provider powers your analytics; switch anytime without code changes. Self-hosted runtime also supported for sovereign deployments.

PLATFORM CAPABILITIES

Features That Unlock Your Data

From natural-language SQL to document RAG — built for teams that need answers fast and auditable.



Text-to-SQL

Ask in plain English or Indonesian. LLM generates executable SQL against your schema, with intent classification to filter out non-data questions and follow-ups.



Knowledge Documents

Upload PDF, Markdown, DOCX, or PPTX as knowledge sources. Ask questions and get answers with inline citations to specific pages. Tesseract OCR fallback for scanned documents (ID + EN).



SQL Self-Correction

When generated SQL fails validation, the system attempts up to three repairs before surfacing an error — fewer dead-end answers, more correct queries.



Knowledge Management

Curated SQL pairs and natural-language instructions steer the LLM toward your team's conventions. Thread-based chat history captures every Q&A, plus recommended follow-up questions based on schema and context.



MDL Semantic Layer

Declarative Modeling Definition Language for tables, columns, relationships, metrics, and cubes. A Rust-based query engine compiles MDL-aware SQL into the right database dialect, with vector-backed retrieval for relevant schema context.



Multi-Dashboard

Pin chart results into multiple saved dashboards per workspace. Star frequently-used dashboards for fast access. Share read-only or edit-capable links with teammates.



Auto-Chart Generation

Picks the right visualization for the data — bar, line, pie, scatter — and renders it through Vega-Lite. No chart configuration needed.



Bilingual ID / EN

End-to-end Indonesian localization — UI strings, LLM responses, and document parsing. OCR language packs for both Indonesian and English scanned PDFs.



REST + GraphQL API

GraphQL for the internal UI, REST under `/api/v1/*` for external integration. Automate provisioning and embed Analytics in your apps.

SUPPORTED DOCUMENT FORMATS

PDF

native text + OCR
fallback

Markdown

heading-based
structure

DOCX

H1/H2/H3
preserved

PPTX

per-slide
structure



USE CASES

Built for Every Data Question

Designed for organizations that need analyst-quality answers without an analyst-sized bottleneck.



Government & Ministries

On-premise deployment for **data sovereignty**. Bilingual Indonesian/English interface. Document RAG for regulations, decrees, and internal SOPs. Connect to internal data warehouses.



Enterprise Self-Service BI

Replace the "ticket the analyst" workflow. Business teams ask in plain language; curated SQL pairs and MDL ensure correctness. Analysts manage the semantic layer and knowledge; everyone else **self-serves answers**.



Knowledge & Compliance Q&A

Upload policy documents, contracts, technical manuals, and standards. Staff ask questions and get **page-cited answers** grounded in your library. OCR support for scanned regulations and historical archives.



Financial & Operational Analytics

Connect to PostgreSQL, Snowflake, BigQuery, ClickHouse, or Trino. Ask cross-source questions; the engine generates correct SQL per dialect. Auto-charts and dashboards for **finance, ops, and management reporting**.

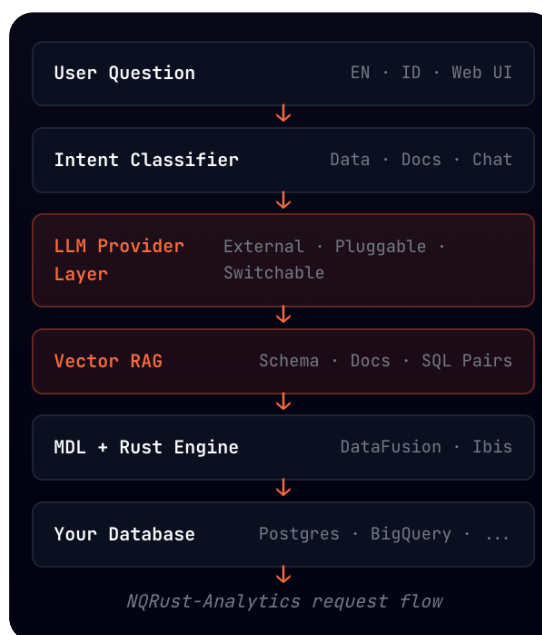
UNDER THE HOOD

Rust engine, semantic layer, **vector retrieval**.

A user's question enters through the chat UI. **Intent classification** routes data questions to the text-to-SQL pipeline and document questions to the knowledge document RAG pipeline — non-data chat is handled directly.

For data questions, the **vector store** retrieves only the schema fragments relevant to the request. The **MDL semantic layer** resolves business terms to columns and tables, and the **DataFusion / Ibis Rust engine** compiles MDL-aware SQL into the target database dialect.

Generated SQL is validated and self-corrected up to three times before execution. Results stream back to the UI for charting (Vega-Lite) and narrative generation. Every LLM call can be optionally traced for cost and latency observability.



WHY NQRUST - ANALYTICS

AI-Powered BI. Sovereign by Default.

Cut the analyst bottleneck, ground answers in your own data, and keep every byte on your infrastructure.

Natural-Language First

Anyone can ask. The system handles intent, schema retrieval, SQL generation, validation, charting, and narrative. Analysts shift from query writers to knowledge curators.

Data Sovereignty & Bilingual

Self-hosted, on-premise deployment. Your warehouse credentials, queries, and documents never leave your network. Full Indonesian localization end-to-end — including OCR.

External LLM, Your Choice

Pre-integrated with multiple external LLM services. You choose which provider powers the analytics; the platform handles the integration. Switch providers anytime without code changes. Self-hosted runtime also supported.

Grounded in Your Data

Every answer traces back to source — SQL queries against your warehouse, citations to specific pages of your documents. MDL semantic layer keeps business definitions consistent across the team.

Ready to put your data to work?

Contact our team for a guided demo, scoping discussion, and on-premise installation plan. Quotes returned within 48 hours.

► LANDING PAGE

<https://analytics.nexusquantum.id/>

► LIVE DEMO

<https://demo.analytics.nexusquantum.id/home>

► DOCUMENTATION

<https://docs.analytics.nexusquantum.id/>